OUTLINE

- Overview & Project Timeline
- Analysis
- Summary & Business Implications

OVERVIEW

- AdventureWorks 2022 dataset (Microsoft):
  - analysed to answer structured questions
  - provide insights for the company.
- Analysis revealed insights into key relationships:
  - global revenue distribution
  - USA regional sales distribution
  - annual leave & bonuses
  - sick leave taken & job type
  - store trading duration & revenue
  - store size, number of employees & revenue.

PROJECT TIMELINE

Plan
Day 1 - am

Clarified project specification
Understanding of database and relationships.

Extract data- SQL
Day 1 - pm

Formulated SQL queries to extract relevant data for each question.

Analyse - python
Day 2 - am

Analysed the data using Python and open source libraries.

Visualise - python
Day 2 - pm

Produced informative visualisations using Python and open source libraries,

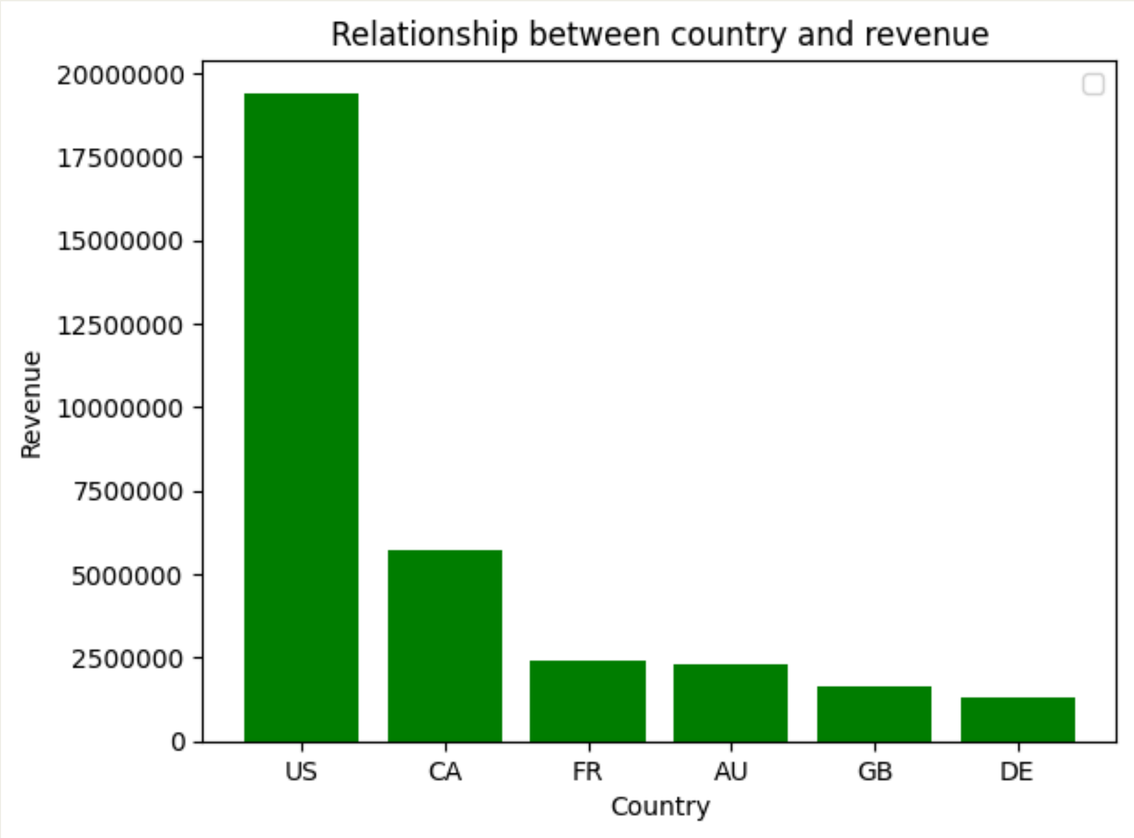
Finalise deliverables
Day 3 - am/pm

Presentation slides.
Project portfolio

DA-LON14

#1 COUNTRY & REVENUE

- USA was the strongest performing country: 19.4 million ‘revenue’ in the last year.
- Canada is the second highest: 5.7 million.
- The 3 European countries and Australia are significantly lower, below 2.5 million.
- The chart shows that the distribution is skewed, with much higher revenues in the USA market compared to others.
- **North American market (USA and Canada) dominated revenue last year. European and Australian markets are much smaller.**
- Profitability, existing competition, demand, and concentration of customers - as well as revenue - required to recommend potential growth opportunities in other countries.



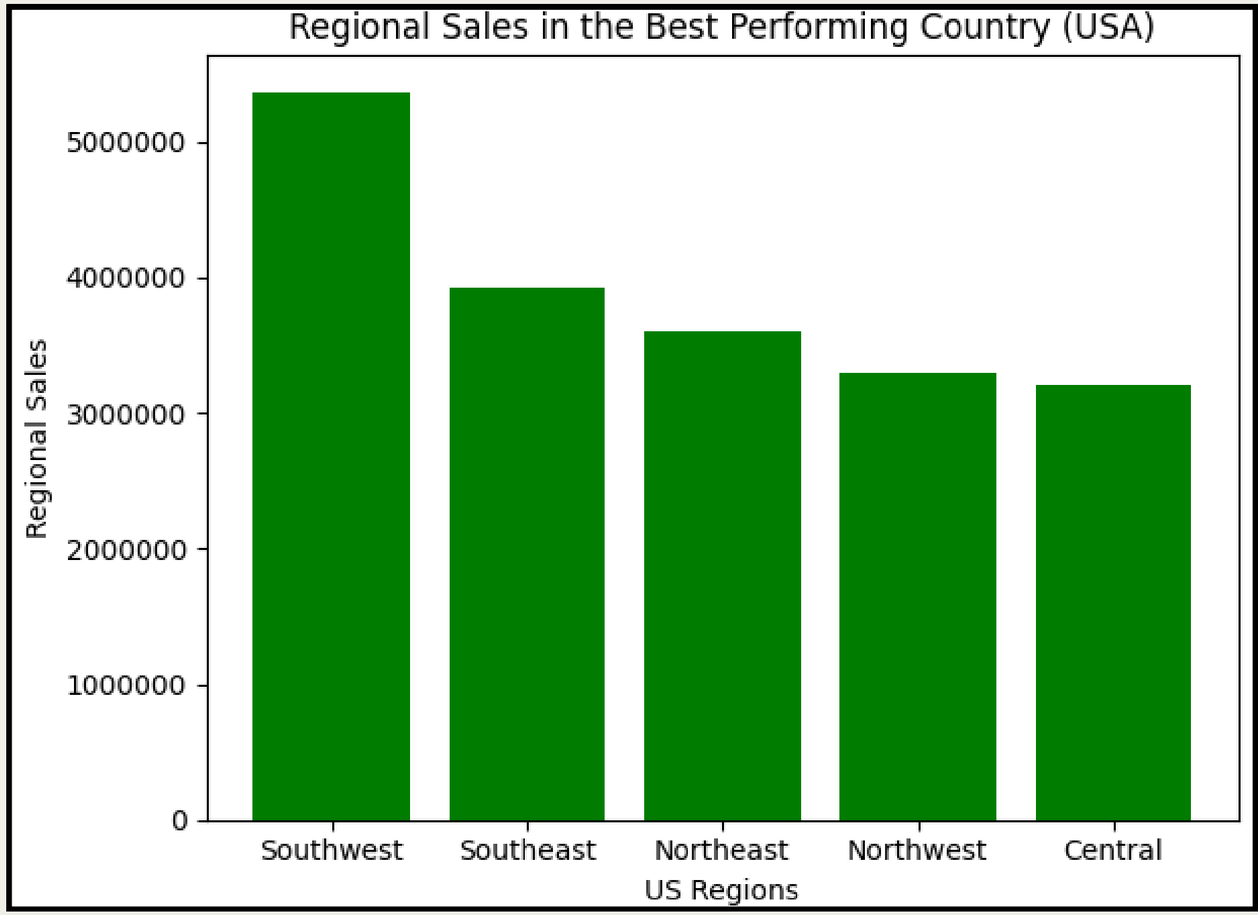
#2 SALES IN THE USA

- **USA:** best performing country (sales) last year.

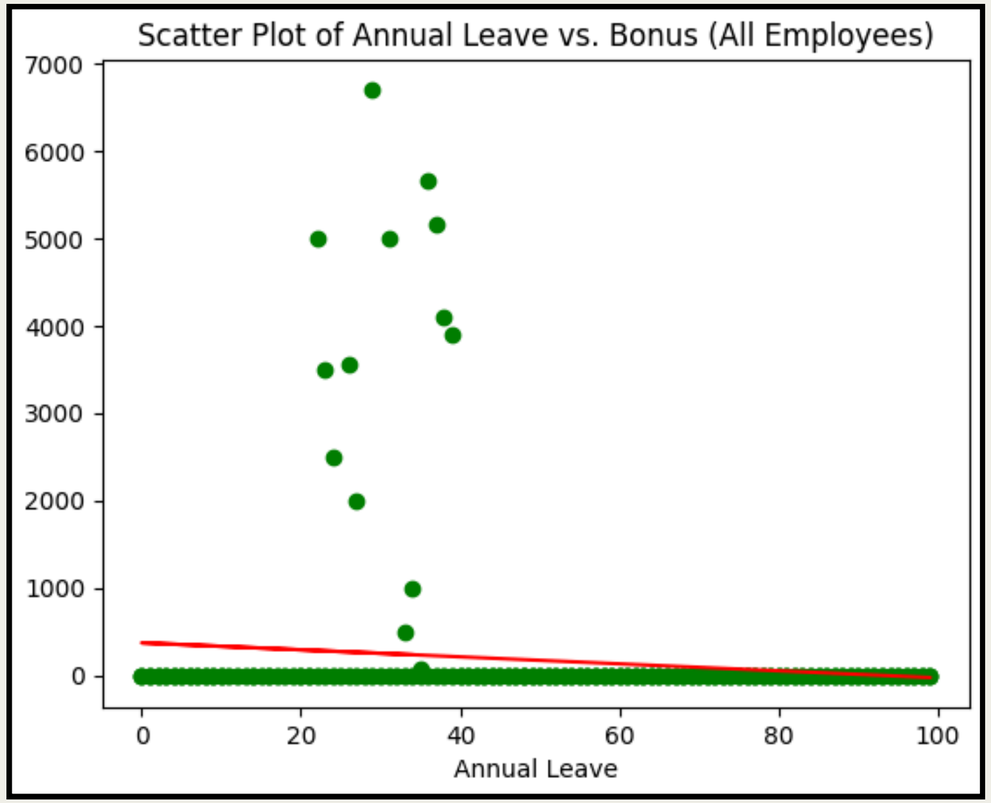
- **Southwest:** best performing US region: 5.4 million sales.

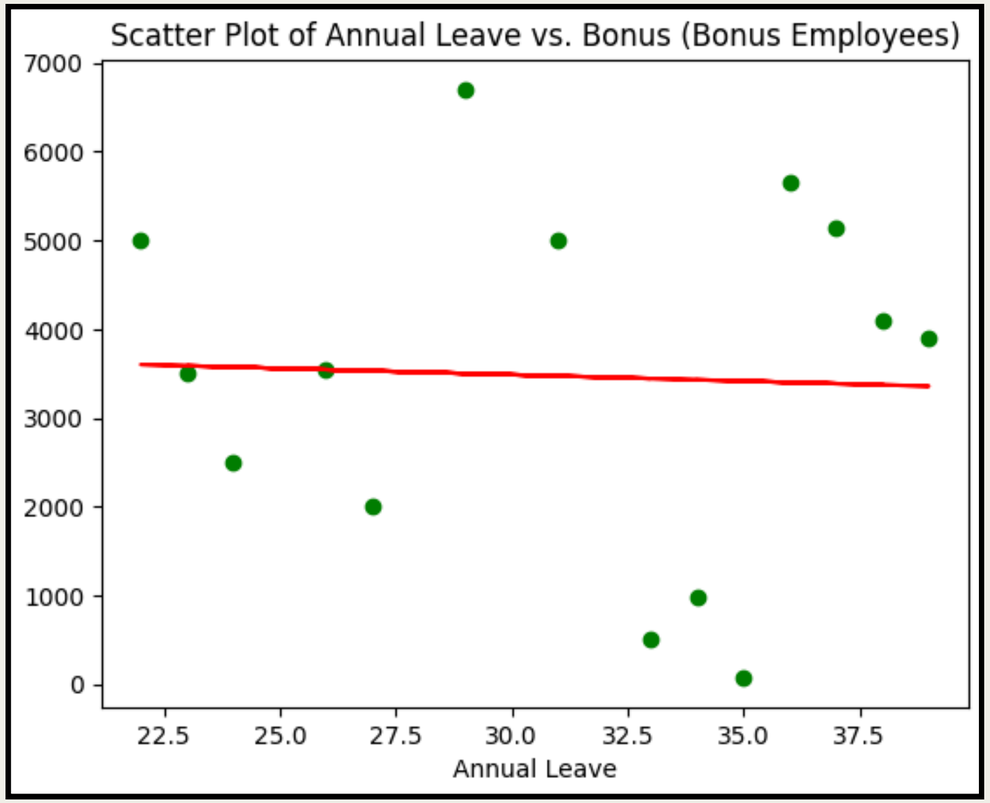
- Bar chart: distribution of US regional sales data for last year:
  - southwest is the clear leader
  - less variation in distribution of sales in the other four regions . More analysis is required to draw insights:
    - number of stores in regions - are lower sales due to lower average sales per store or are lower sales are due to less stores?
    - impact of seasonal variation in products sold, which may have been less in the southwest.

- Profitability, existing competition, demand, concentration of customers, and seasonality effects: **additional data required to recommend potential growth opportunities in other regions of US.**



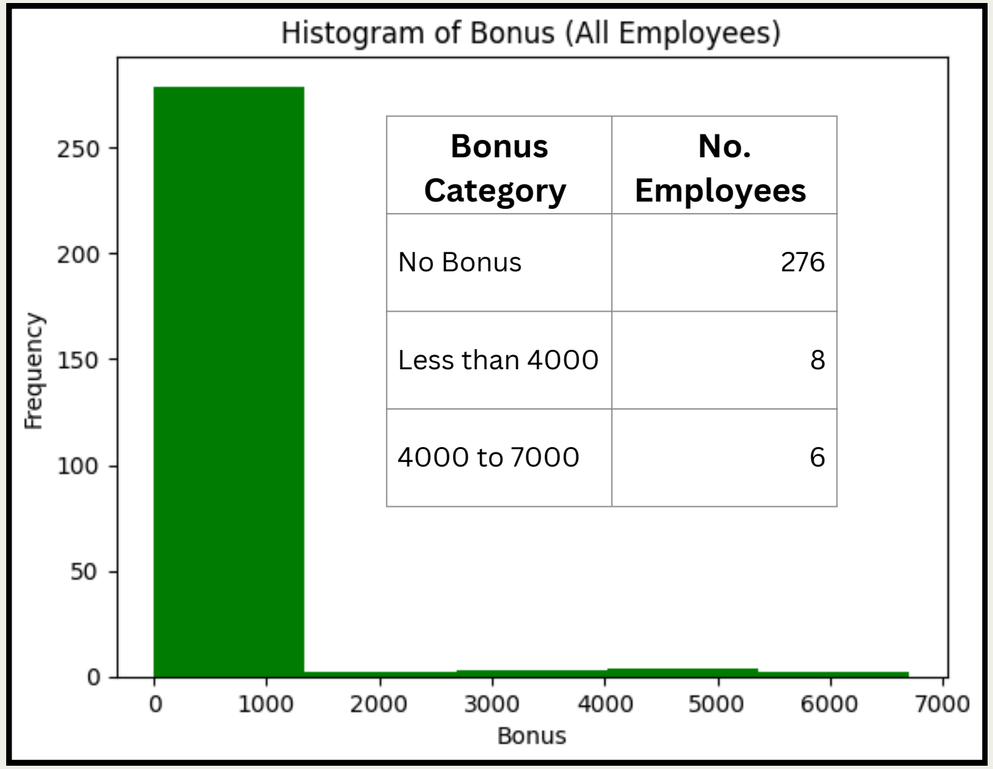
#3 ANNUAL LEAVE & BONUS



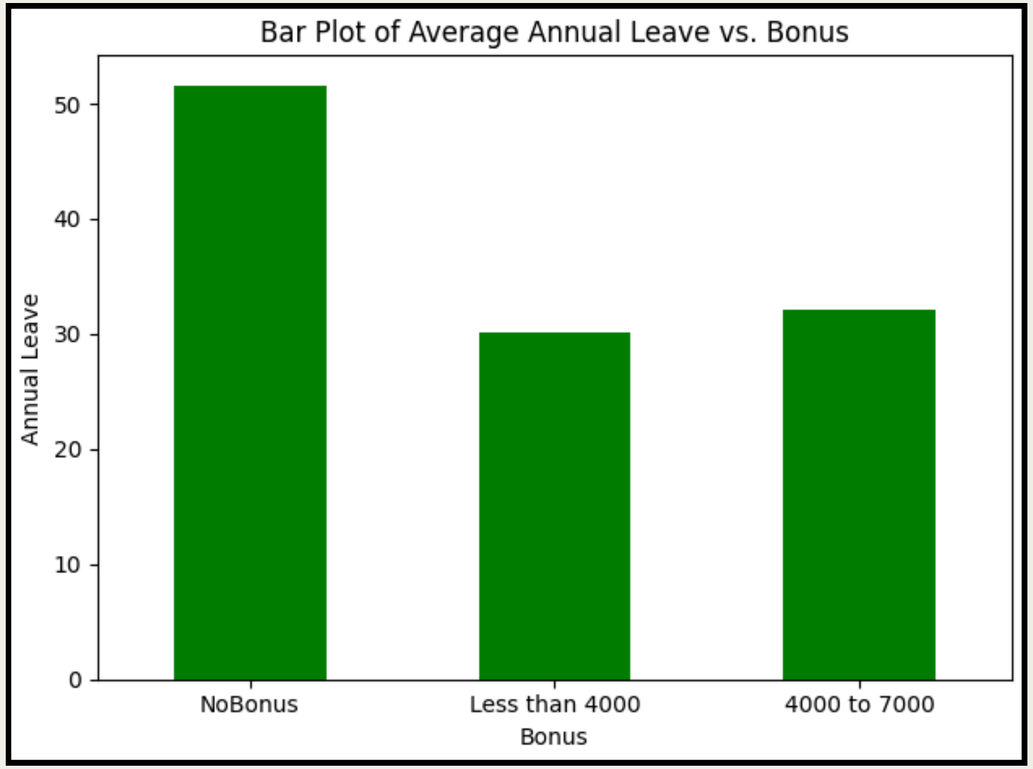


Employees who received a bonus cluster at the lower end of the annual leave distribution.

#3 ANNUAL LEAVE & BONUS

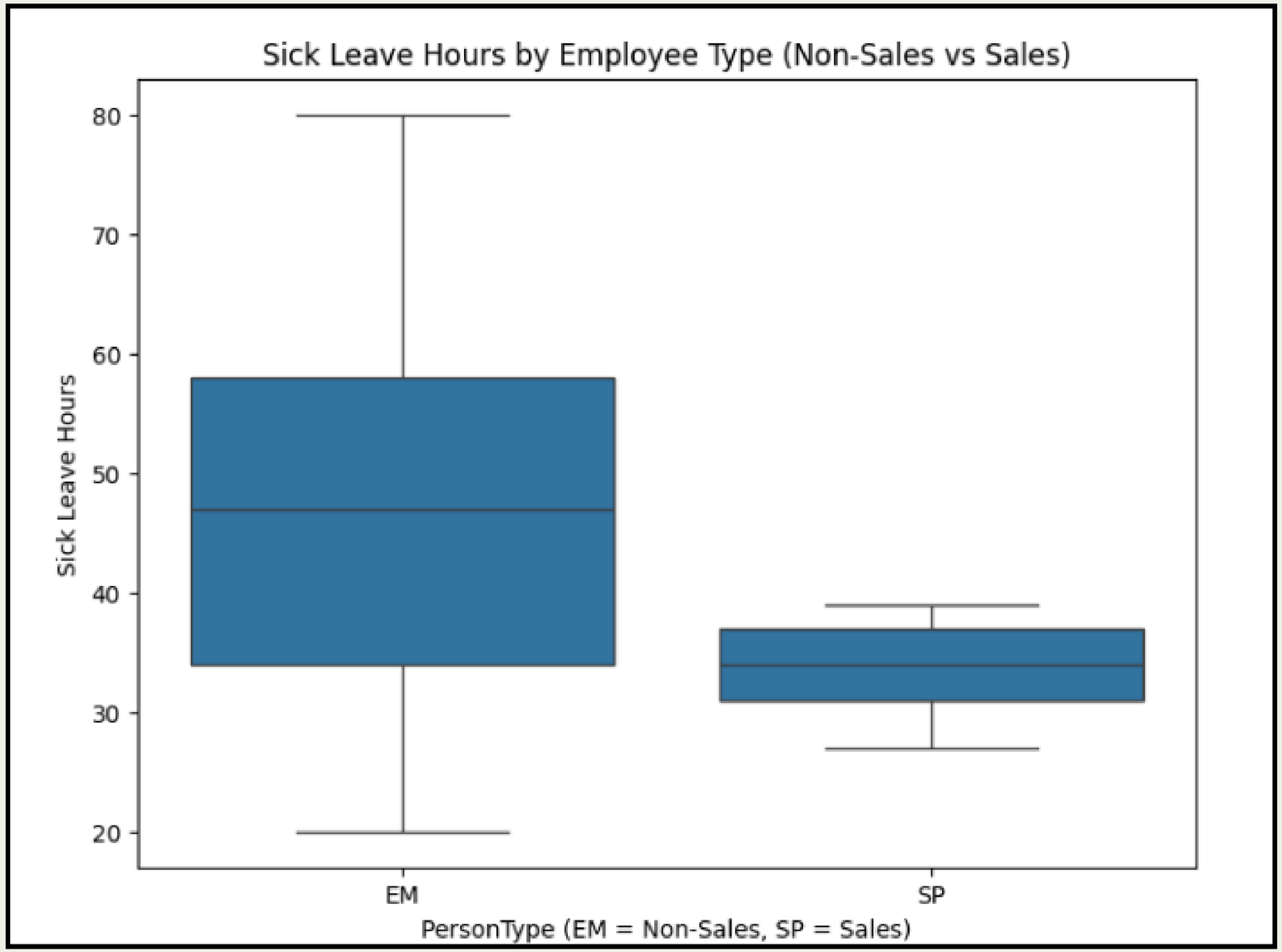


Bonus Category	No. Employees
No Bonus	276
Less than 4000	8
4000 to 7000	6



Employees who received no bonus took a higher average amount of sick leave

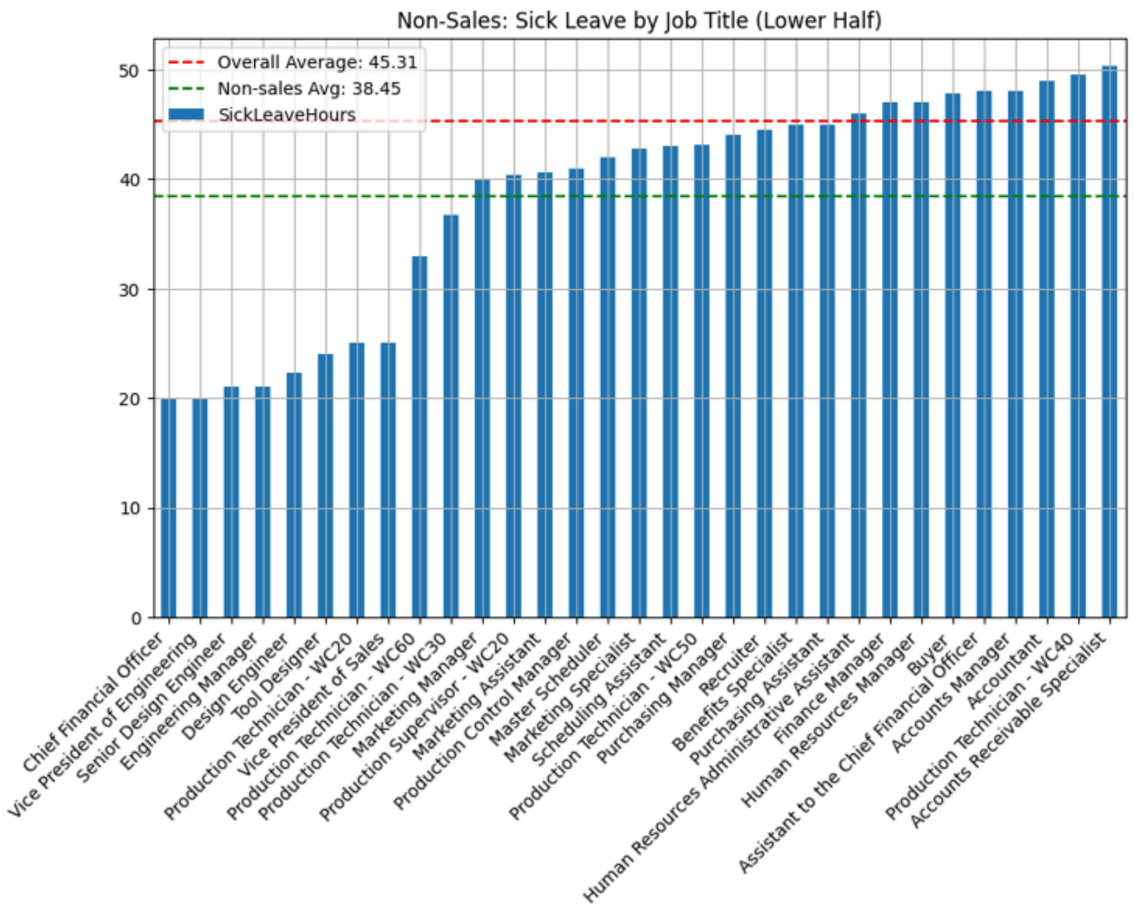
#4 SICK LEAVE & JOB TITLE

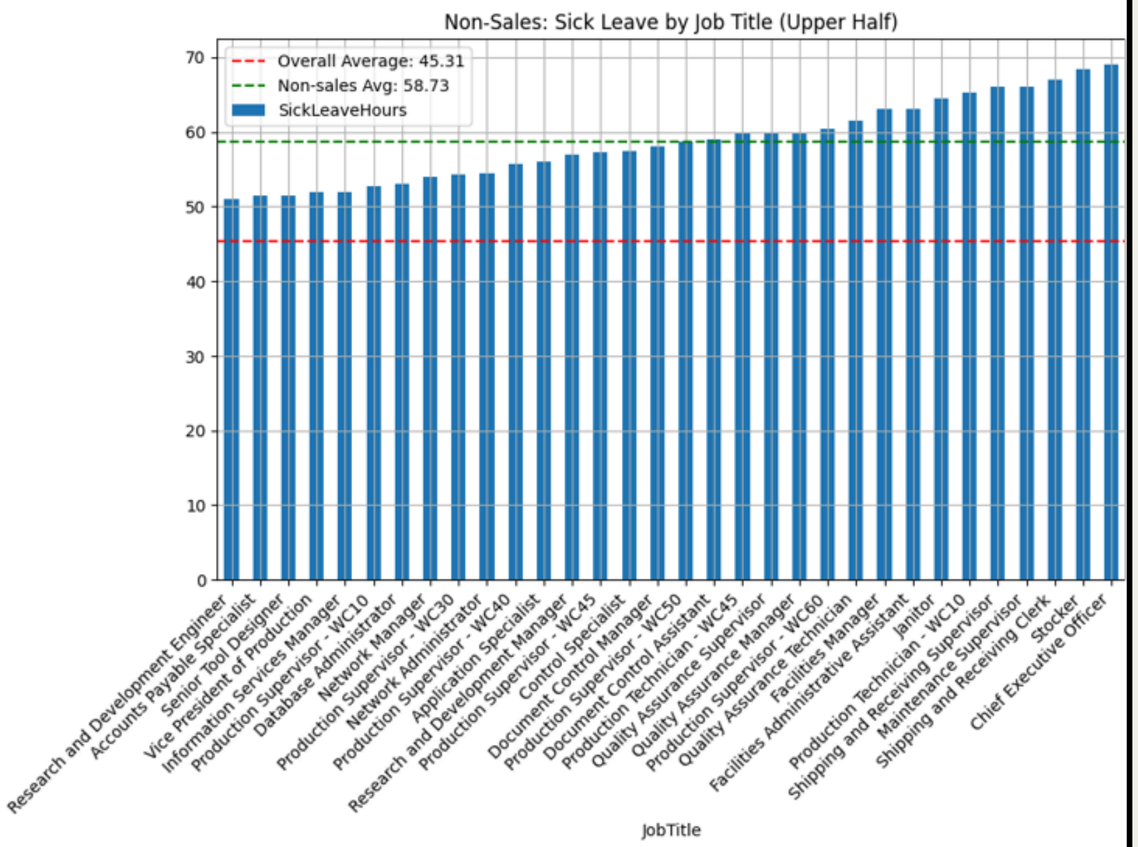


- **Non sales employees took more sick leave hours. Wider variance: 20-80 hours.**
- **Sales employees took less sick leave - similar values, average around 30 hours.**
- **More data is needed for: analysis of tenure and departments.**

#4 SICK LEAVE & JOB TITLE

Two graphs representing two halves of Non-sales EE sick leave data





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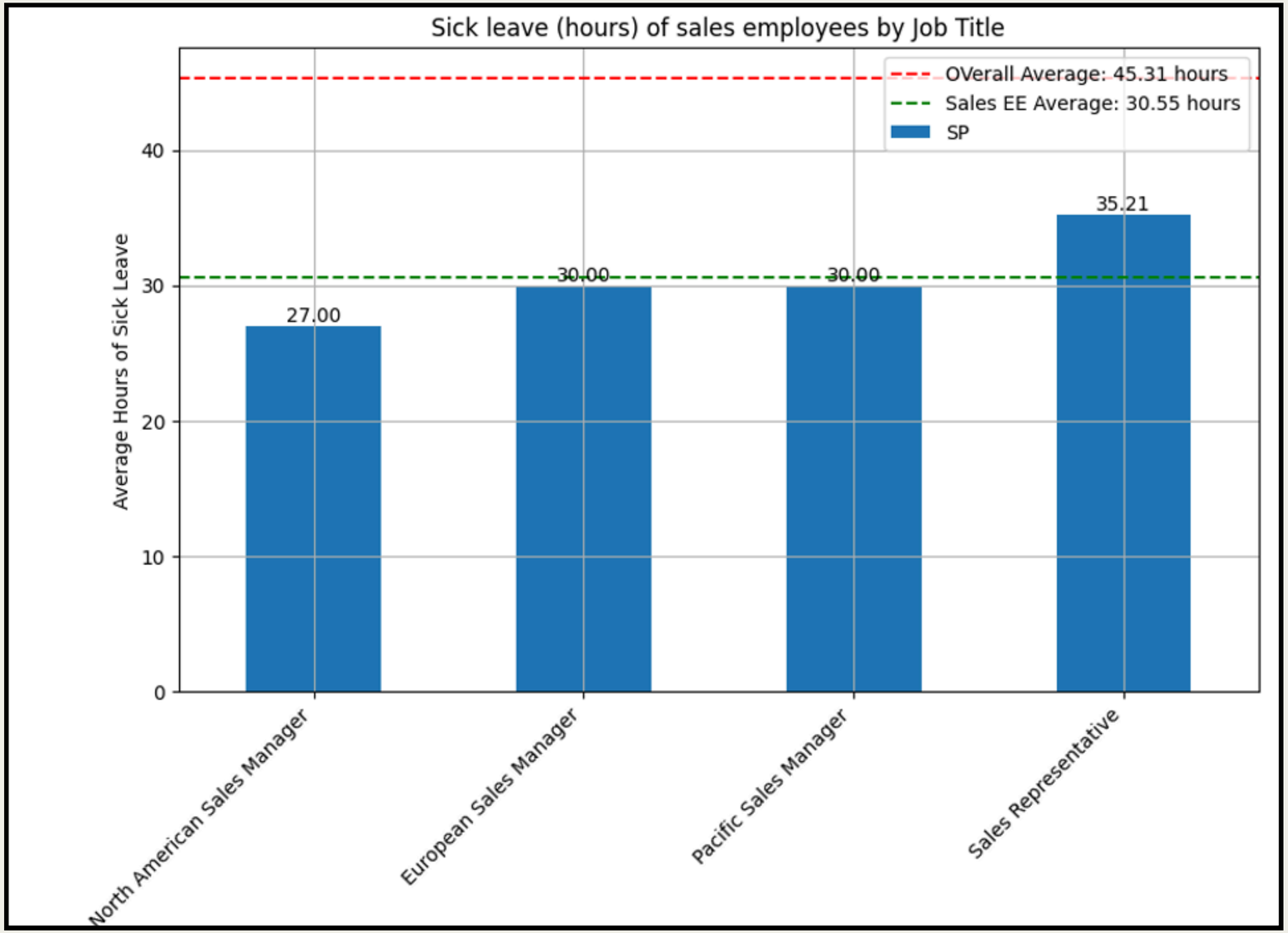
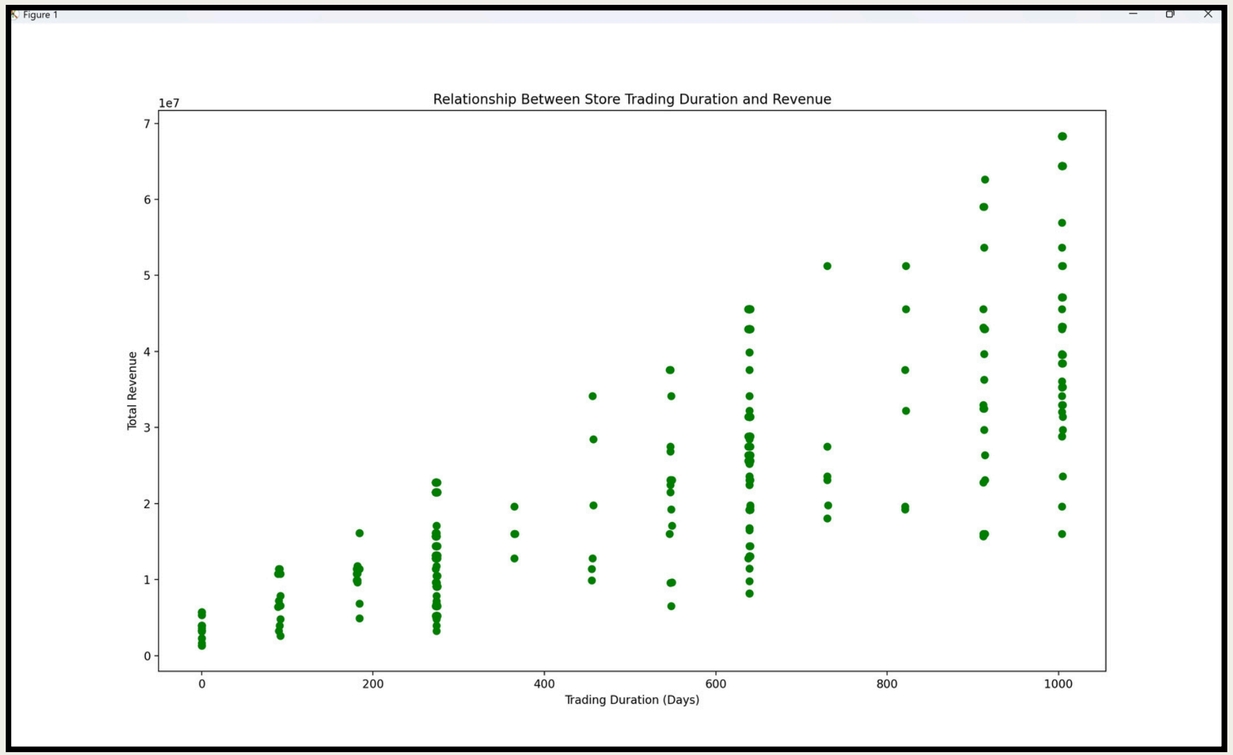
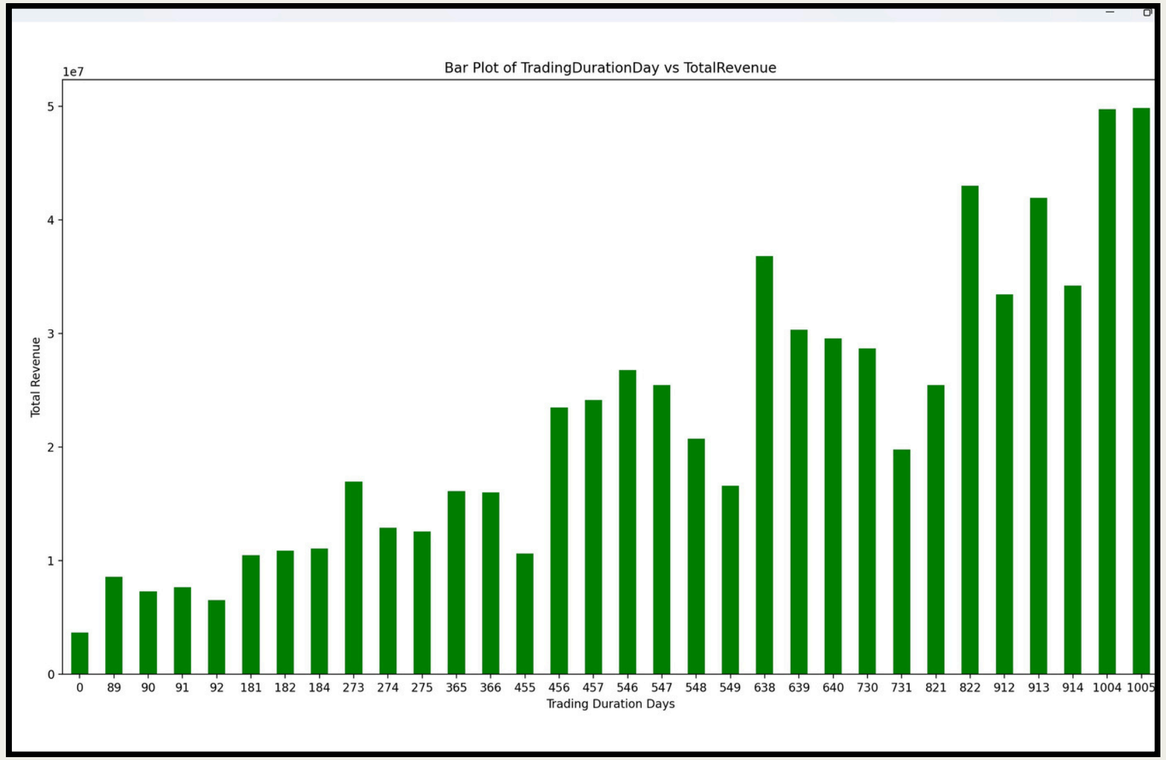


- Non sales people's sick leave is varied - CEO v CFO
- Amongst sales people, sick leave hours is more consistent regardless of job type.

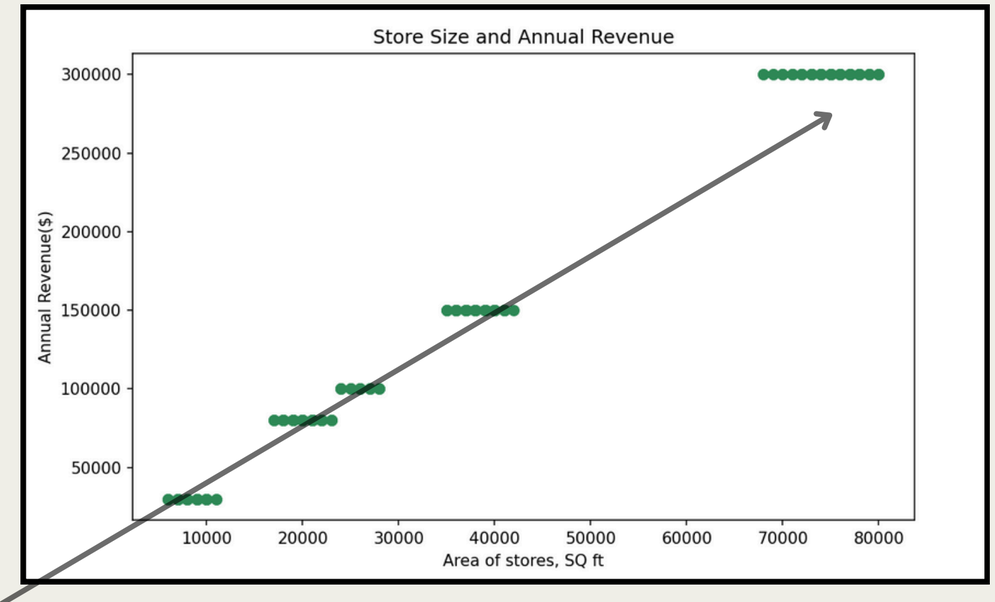
#5 STORE TRADING DURATION & REVENUE

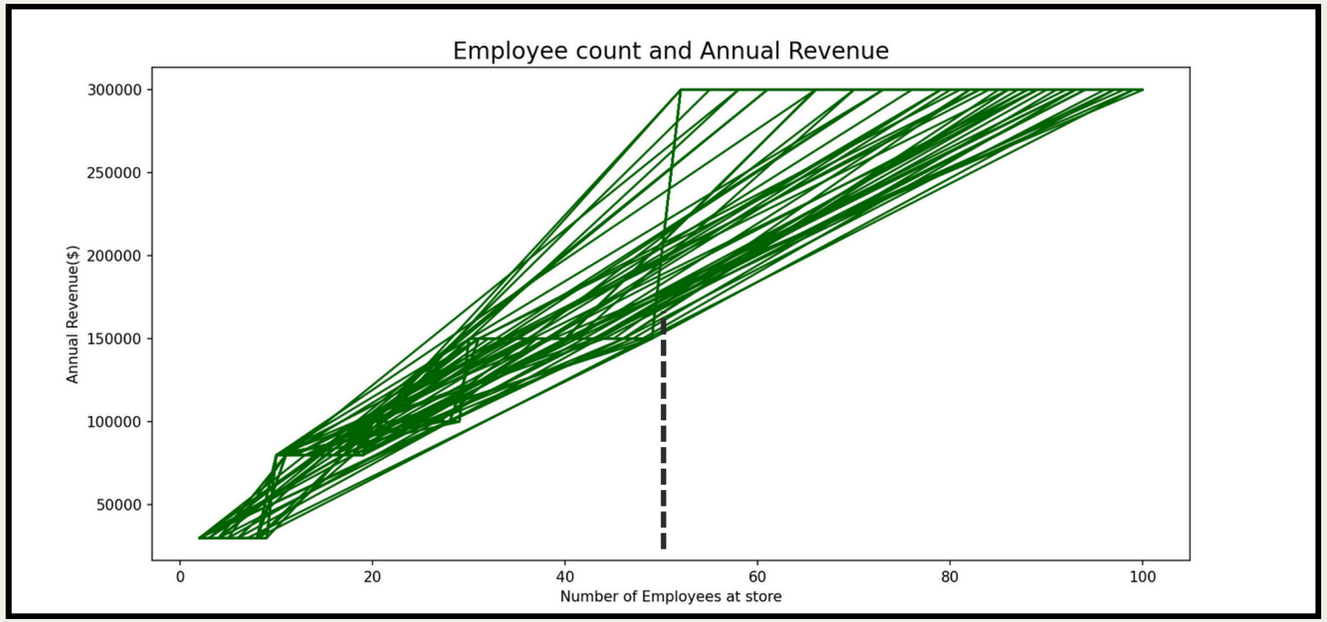




STORES WITH LONGER TRADING DURATIONS TEND TO GENERATE HIGHER REVENUE OVERALL.

#6 STORE SIZE, EMPLOYEE NO, & REVENUE





- **Increased revenue can be linked to larger stores**
- **No large outliers**
- **Need to investigate further and find how the type of store or location can benefit annual revenue**

- **Generally more revenue generated in stores with more employees**
- **Stores with lower staff numbers are concentrated in low revenue range**
- **Significantly higher revenue with at least 50 employees**

BUSINESS IMPLICATIONS OF DATA ANALYSIS

- Profitability and competition data - as well as revenue - is needed to consider *potential for growth opportunities in European and Australian markets.*

- Additional analysis is needed to understand *why southwest is the strongest sales region in USA:*
  - Seasonality of products sold influences sales markets in different regions of the USA; any seasonality effect may not be as strong in the southwest.
  - Variation in store numbers / concentration of potential customers?

- Additional analysis/metrics are needed to draw insights between annual leave and bonuses, and *how the company's incentive structure can be improved.*

BUSINESS IMPLICATIONS

- Further analysis of non-sales people is needed to understand the drivers of sick leave, *to design effective wellbeing policies.*

- The positive reputation of more established stores influences increased revenue. *But non profitable stores may have closed down and are not included in recent data.*

- Investing in larger stores and hiring more employees may continue to promote higher revenue; *however, profitability needs to be considered too.*